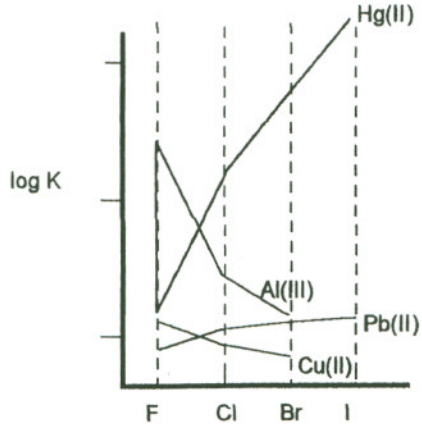
INORGANIC CUME
APRIL 26, 2002

ACID/BASE CHEMISTRY

1 - Use the HSAB model to briefly account for the trends in log $K_{formation}$ for the Lewis acid-base adducts formed by Hg^{2+}, Al^{3+}, Cu^{2+} and Pb^{2+} with the halide anions shown in the graph below.



2 - Pauling proposed an empirical rule for estimating the pKa for oxoacids [pKa ~ 8 – 5p + 5z for $H_nXO_{n+p}^{-z}$]. Qualitatively, justify the signs of the p and z terms.

Answer **two** of 3 – 5

3 - Suggest a reason to account for the breakdown of the Pauling rule for consecutive pK's of acids such as $H_4S_2O_7$. [i.e. $pK_1 \sim pK_2 << pK_3$]

4 - Suggest a rationale for the observation that pK_1 values for H_3PO_3 (~ 1.8) and H_3PO_2 (~2) are much smaller than predicted by the Pauling rule.

5 - Two dramatically different values are often quoted for pK_1 for carbonic acid (3.6 and 6.3). Why are both values "correct"?

Answer **one** of 6 – 8

6 - Predict the order of acid strength for the molecules $H(X)SO_2$ [- X = - CF_3, - NH_2, - OH, - F] and qualitatively account for your order based on the nature of the – X group.

7 - The catalytic properties of SiO_2 (s) is often related to the presence of acidic and basic surface sites. Give a molecular description of the difference between these two locations on a silica surface.

8 - Why is the order of base strength often different for the same set of molecules in aqueous solution and the gas phase?